Abstract

Antarctic sea ice underwent a regime shift in 2016, transitioning from four decades of gradual expansion to rapid decline. Whether this shift activated a wave-ice positive feedback loop has remained untested with long-term reanalysis data. Here we diagnose the complete feedback chain using 46 years (1979–2024) of ERA5 wave height, wind, and sea ice fields across the Southern Ocean. Sea ice loss opens more water at the ice edge, yet this additional open water does not generate larger waves locally. Instead, sea ice independently modulates wave height by controlling how much remotely generated swell survives the transit across the ice pack. This effect is strongest in austral winter, when ice coverage is maximum, and persists after accounting for wind variability. The wave energy flux at the ice edge barely changed between the pre-2016 and post-2016 periods (+2.4%); rather, the ice edge retreated poleward into a region of pre-existing, wind-driven wave energy. The mean wave period at the ice edge (9.0 s) confirms that swell, not locally generated wind-sea, dominates. These results identify a seasonally mediated, swell-attenuation mechanism linking sea ice loss to enhanced wave exposure: the ice retreated into the waves, rather than the waves getting bigger.

Antarctic sea ice retreated into pre-existing waves: swell attenuation, not fetch, controls wave exposure at the ice edge

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June 14, 2026

1 Introduction

Antarctic sea ice extent increased gradually from 1979 to 2014, then declined precipitously. The annual mean extent dropped by 11.5% between the pre-2016 and post-2016 periods, with the Pettitt change point test placing the regime shift at 2016 ($p = 0.004$)[1]. The decline was most pronounced in austral summer (December–February), when extent fell by 16.8%. This abrupt transition, unprecedented in the satellite era[2–4], has prompted the question of whether the Southern Ocean has crossed a tipping point into a new, self-reinforcing state of ice loss[5].

A leading hypothesis invokes a wave-ice positive feedback. Sea ice acts as a buffer that attenuates ocean swell before it reaches ice shelf margins and the interior ice pack[6, 7]. When ice retreats, the open-water fetch lengthens, wave energy increases, and stronger waves fragment and erode the marginal ice zone (MIZ), exposing yet more open water. This feedback loop, if active, would accelerate ice loss beyond what atmospheric forcing alone could produce and would have direct implications for ice shelf stability, since Massom et al.[6] demonstrated that ocean swell triggered the disintegration of the Larsen A, Larsen B, and Wilkins ice shelves following loss of their protective sea ice buffer.

The individual links of this chain have theoretical and observational support. Wave-ice interaction theory predicts that sea ice acts as a low-pass filter, preferentially attenuating short-period waves, with attenuation rates depending on ice concentration, thickness, and floe size[7–9]. Wave-triggered breakup generates lognormal floe size distributions in the MIZ[10], and fractured ice melts faster, creating a local feedback between fragmentation and concentration loss. Recent observations confirm that ocean waves drive Antarctic sea ice melting through flooding and mechanical pulverization[11], and that waves can propagate over 1000 km into the winter ice pack[12]. ICESat-2 altimetry has enabled direct observation of wave attenuation through the Antarctic MIZ[13].

Independently, Southern Ocean wave climate has been changing. Satellite altimetry and reanalysis products show significant increases in significant wave height (SWH) across the Southern Ocean over recent decades, driven primarily by the intensification of the circumpolar westerlies associated with the positive trend in the Southern Annular Mode (SAM)[14]. Young and Ribal[15] reported global wave height trends from 33 years of satellite altimetry, finding the strongest increases at high southern latitudes. These wind-driven wave changes provide the background energy field against which any wave-ice feedback must operate.

What remains untested is whether the individual wave-ice interaction mechanisms close into a self-reinforcing loop at the basin scale, and specifically whether the 2016 regime shift activated or strengthened such a feedback. Two competing hypotheses can be distinguished (Figure 1a).

The *fetch hypothesis* posits that ice retreat exposes more open water, lengthening the distance over which winds can generate waves, thereby increasing SWH through local wave growth. The *swell-attenuation hypothesis* posits that ice retreat reduces the path length over which remotely generated swell is attenuated before reaching the ice edge, thereby increasing SWH without additional local wave generation. These two mechanisms predict different wave spectral signatures: the fetch mechanism would increase short-period wind-sea, while the swell-attenuation mechanism would increase long-period swell.

Addressing this question requires simultaneous, long-term observations of sea ice concentration, wave height, wind forcing, and MIZ geometry across the full Southern Ocean (Figure 1b), a combination that only reanalysis products can provide over multi-decadal timescales. Here we use 46 years (1979–2024) of ERA5 monthly reanalysis to diagnose the complete wave-ice feedback chain. We test whether sea ice loss increases fetch at the ice edge, whether increased fetch or reduced sea ice amplifies SWH, whether higher SWH drives ice-edge retreat, and whether ice-edge retreat feeds back to reduce total sea ice concentration. We apply Granger causality with Bonferroni correction, partial Granger causality conditioning on wind speed to isolate the independent effect of sea ice, and seasonal decomposition to identify the seasons when wave-ice coupling is strongest. We also assess changes in ice shelf buffer conditions at five major Antarctic ice shelves.

2 Data and Methods

2.1 ERA5 reanalysis products

We use three ERA5 monthly-mean reanalysis products from the European Centre for Medium-Range Weather Forecasts (ECMWF)[16]:

- **Significant wave height (SWH)**, mean wave period (MWP), and mean wave direction (MWD) on the native 0.5° wave model grid, 1979–2024 (552 months). The ERA5 wave model (WAM) runs only in grid cells where sea ice concentration is below 30%, so SWH data are inherently limited to open and marginal waters.
- **10-m wind components** (u_{10}, v_{10}) on the 0.25° atmospheric grid, 1979–2024.
- **Sea ice concentration** (SIC) on the 0.25° atmospheric grid, 1979–2024. ERA5 SIC is derived from OSTIA satellite observations (from 1979 onward) assimilated into the ECMWF system.

All fields are restricted to the Southern Ocean domain 40°S–75°S, all longitudes. Wind speed is computed as $|\mathbf{u}_{10}| = \sqrt{u_{10}^2 + v_{10}^2}$.

2.2 NSIDC Sea Ice Index

We use the NSIDC Sea Ice Index v4.0[17] as an independent validation of Antarctic sea ice extent trends. Monthly Southern Hemisphere extent and area are available from 1979 to 2026. These satellite-derived estimates are independent of ERA5 SIC and provide a check on the timing and magnitude of the 2016 regime shift.

2.3 Regional definitions

We define four latitude bands for area-weighted spatial averaging:

- **Full Southern Ocean**: 40°S–75°S
- **Near-MIZ zone**: 55°S–65°S (captures the mean position of the seasonal ice edge and MIZ)
- **ACC belt**: 45°S–55°S (largely ice-free, dominated by the Antarctic Circumpolar Current)
- **Deep ice edge**: 65°S–75°S (persistently ice-covered in winter)

The **marginal ice zone (MIZ)** is defined as the band where $0.15 < \text{SIC} < 0.80$. MIZ width is computed as the zonal-mean meridional extent of this band at each time step.

2.4 Ice-edge-based fetch and wave height sampling

For each month, the ice edge is defined as the most equatorward latitude at which SIC exceeds 0.15 at each longitude. The zonal-mean ice-edge latitude serves as a proxy for fetch: a more poleward ice edge implies a longer meridional distance of open water available for wave generation by the prevailing westerlies. This ice-edge-based fetch captures the dominant zonal-wind geometry of the Southern Ocean, where fetch is limited primarily by the meridional position of the ice margin rather than by zonal extent.

SWH is sampled at the ice edge rather than at fixed latitude bands, to avoid conflating changes in ice-edge position with changes in wave climate. For each longitude, SWH is extracted from the nearest grid point 2° equatorward of the local ice edge, ensuring that the sampled wave height reflects conditions actually experienced by the ice margin.

2.5 Change point detection

We apply the Pettitt test[18] to annual-mean anomaly time series of each variable to detect regime shifts. The Pettitt test is a non-parametric test for a single change point in the mean of a time series. The test statistic K and associated p-value are computed following the standard formulation. We report change point years, pre- and post-shift means, and significance levels.

2.6 Granger causality

We test directional predictability between pairs of monthly anomaly time series using Granger causality[19]. For each pair (x, y), we test whether past values of x improve the prediction of y beyond what y's own past values provide. We evaluate lag orders 1–6 months, selecting the optimal lag by AIC. We report the F-statistic and p-value of the restricted-vs-unrestricted model comparison.

The feedback chain is tested as a sequence of Granger causality links:

$$\text{SIC} \rightarrow \text{Fetch} \rightarrow \text{SWH} \rightarrow \text{Ice edge} \rightarrow \text{SIC}$$

with wind speed ($|\mathbf{u}_{10}|$) as an alternative driver of SWH. To address the multiple comparisons problem, we report Bonferroni-corrected p-values alongside uncorrected values for all six tests conducted simultaneously.

2.7 Partial Granger causality

To distinguish the independent effects of sea ice and wind on SWH, we apply partial Granger causality. For the test "does SIC Granger-cause SWH after controlling for wind?", the full model includes lagged values of SWH, SIC, and wind speed, while the reduced model omits the lagged SIC terms. The F-test then assesses whether SIC adds predictive power beyond what SWH's own history and wind already provide. This controls for the possibility that both SIC and SWH are driven by a common atmospheric forcing (e.g., SAM) with different lag structures.

2.8 Seasonal decomposition

We test the seasonal dependence of the SIC→SWH Granger link by restricting the analysis to months within each austral season: DJF (summer), MAM (autumn), JJA (winter), and SON (spring). This tests the physical prediction that the swell-attenuation mechanism should be strongest in winter, when sea ice coverage is maximum and the attenuation path is longest. For seasonal tests, the maximum lag is reduced to 3 months to preserve statistical power within each 138-month seasonal subsample.

2.9 Ice shelf buffer analysis

For five major Antarctic ice shelves (Ross, Filchner-Ronne, Amery, Larsen C, and Totten), we compute the number of months per year in which the mean SIC in a box encompassing the shelf front exceeds 0.15 ("buffer months"). A decrease in buffer months indicates increased exposure of the ice shelf front to open-ocean wave action.

3 Results

3.1 Confirmation of the 2016 sea ice regime shift

The NSIDC Sea Ice Index confirms a significant regime shift in 2016 (Pettitt $p = 0.004$; Figure 3a), with annual mean extent declining from 11.68 to 10.33 million km^2 (-11.5%). The decline is strongest in austral summer (DJF: -16.8%, $p = 0.002$) and weakest in winter (JJA: -5.7%, not significant at $p = 0.073$; Figure 3b). The ERA5 SIC field confirms this timing: the area-weighted mean SIC in the ice zone (55°–75°S) shows a change point at 2016 ($p = 0.021$), and the open-water fraction shifts synchronously ($p = 0.013$). Before 2016, Antarctic sea ice extent increased at $+0.022$ million km^2 yr^{-1} ($p < 0.001$); after 2016, it declined at -0.311 million km^2 yr^{-1} ($p = 0.054$, marginally significant over only 9 years of post-shift data).

Spatially, the sea ice loss is not uniform around Antarctica (Figure 2a). The largest SIC reductions occur in the Bellingshausen and Amundsen Sea sectors (West Antarctica), with secondary losses in the Weddell Sea and East Antarctic sectors. The ice edge retreated poleward across most longitudes (Figure 2c), but the retreat is most pronounced in West Antarctica where it exceeds 2° of latitude. The ice-edge latitude and zonal-mean fetch proxy both show Pettitt change points synchronous with the extent shift at 2016 ($p = 0.009$; Figure 5a,d), confirming that the regime shift manifests in both the total extent and the spatial position of the ice margin.

3.2 Wave height trends are wind-driven, not fetch-driven

Significant wave height in the Southern Ocean has increased over the satellite era (Figure 4a), but the change point occurs in 1993 ($p = 0.001$), more than two decades before the sea ice regime shift. In the near-MIZ zone (55°–65°S), the SWH anomaly shows a marginal change point at 1993 ($p = 0.055$), with mean anomaly shifting from -0.078 m to $+0.034$ m. The ACC belt (45°–55°S) shows the strongest SWH increase ($p = 0.0001$), consistent with intensification of the westerly wind belt. Spatially, 17% of Southern Ocean grid points show a significant SWH increase between the pre-2016 and post-2016 periods ($p < 0.05$, two-sample t-test).

The correlation between near-MIZ SWH anomaly and sea ice extent anomaly is statistically significant but weak ($r = 0.136$, $p = 0.001$, $n = 550$ months), indicating that sea ice conditions explain less than 2% of SWH variance.

3.3 Does ice loss change waves, and through what mechanism?

Granger causality tests reveal which links in the hypothesized feedback chain hold and which fail (Table 1):

After Bonferroni correction for six simultaneous tests (Table 1; Figure 5e), only the SIC→Fetch link remains significant at the 5% level. The remaining links are individually significant ($p < 0.05$) but marginal after correction ($p_{\mathrm{Bonf}} = 0.11$–0.30). The SIC→Fetch relationship is robust across sub-periods: it holds in both the pre-2016 ($F = 16.0$, $p < 0.001$, $n = 444$ months) and post-2016 ($F = 21.1$, $p < 0.001$, $n = 108$ months) periods, with the post-2016 effect being stronger.

The direct SIC→SWH link, though marginal after correction on annual data, exhibits strong seasonal dependence. When tested separately by season (Figure 6a), SIC Granger-causes SWH only in austral winter (JJA: $F = 5.10$, $p = 0.007$; Bonferroni-corrected for 4 seasonal tests: $p_{\mathrm{Bonf}} = 0.028$), with no significant relationship in other seasons (DJF $p = 0.94$, MAM $p = 0.28$,

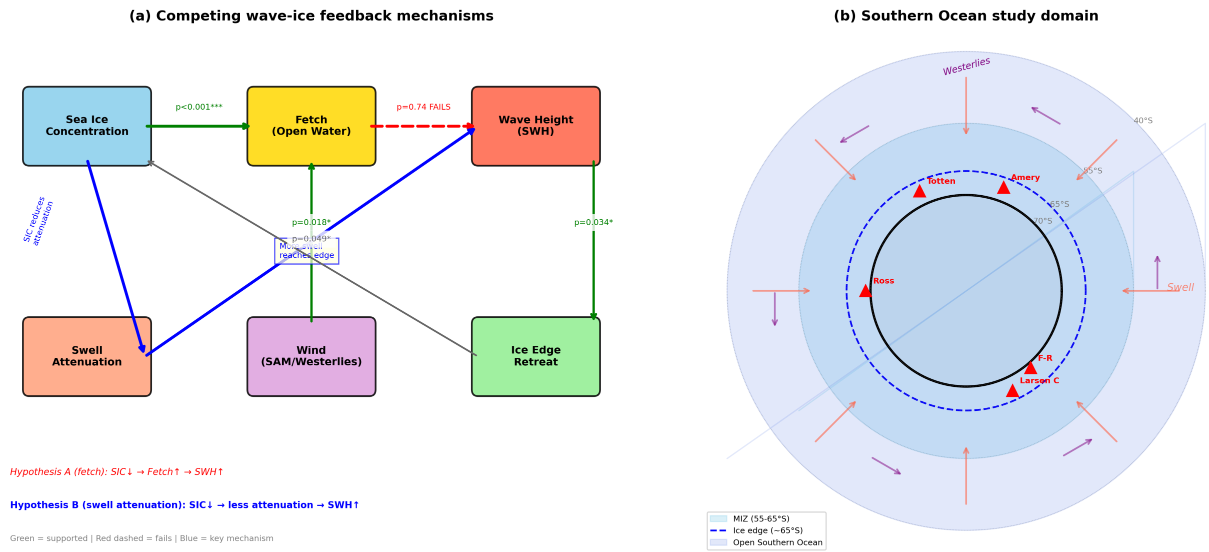

Figure 1: **Competing wave-ice feedback mechanisms and study domain.** **a**, Conceptual diagram of the two hypothesized pathways by which sea ice loss could amplify wave height at the ice edge. Hypothesis A (red, dashed) invokes fetch lengthening: less ice exposes more open water for local wave generation. Hypothesis B (blue, solid) invokes swell-attenuation reduction: less ice shortens the path over which remotely generated swell is attenuated. Green arrows indicate statistically supported Granger-causal links; red dashed arrow indicates the failing link. Granger p-values shown. **b**, Southern Ocean study domain in south-polar stereographic projection. Light blue band: marginal ice zone (55–65°S). Blue dashed circle: mean ice edge (∼65°S). Red triangles: five major ice shelves examined for buffer loss. Red arrows: poleward swell propagation. Purple arrows: circumpolar westerlies.

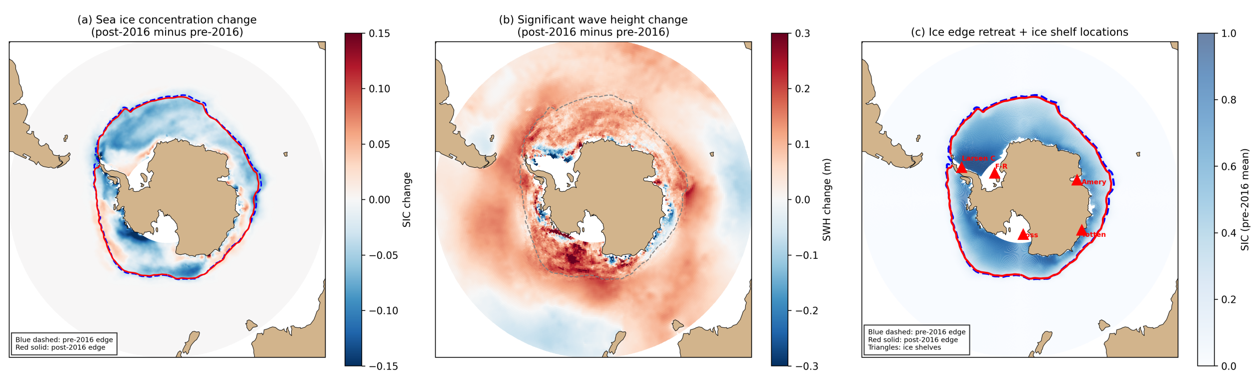

Figure 2: **Spatial patterns of sea ice and wave height change (post-2016 minus pre-2016).** **a**, Sea ice concentration change. Blue dashed contour: pre-2016 ice edge (SIC = 0.15); red solid contour: post-2016 ice edge. Sea ice loss is widespread, with the largest reductions in the Bellingshausen, Amundsen, and Weddell sectors. **b**, Significant wave height change. Increases are concentrated in the ACC belt and near the retreated ice edge. Gray dashed contour: pre-2016 ice edge for reference. **c**, Pre-2016 mean SIC (shading) with pre-2016 (blue dashed) and post-2016 (red solid) ice edge contours overlaid. Red triangles mark the five ice shelves. The poleward retreat of the ice edge is evident across most sectors. All panels use south-polar stereographic projection with Natural Earth coastlines.

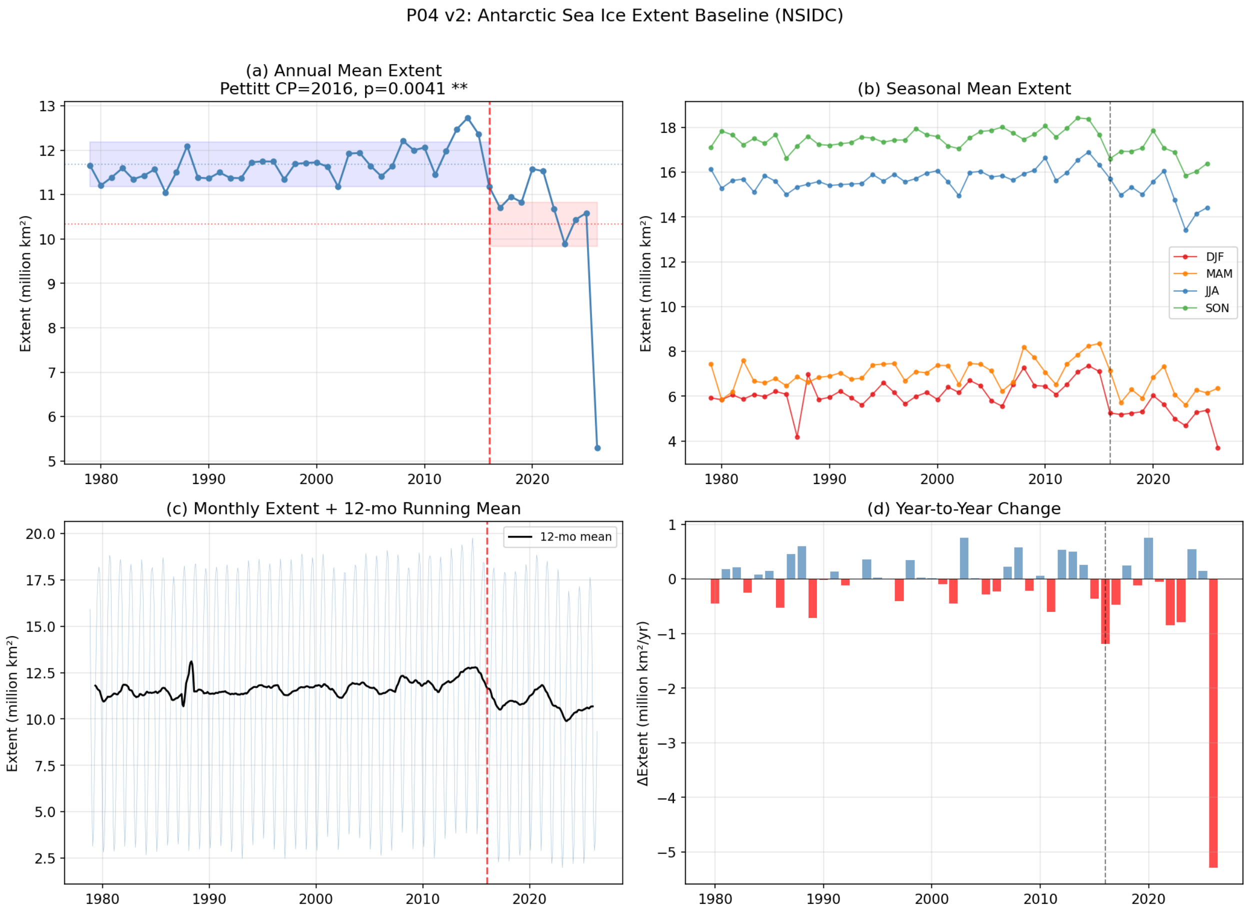

Figure 3: **Antarctic sea ice extent time series (NSIDC Sea Ice Index, 1979–2026). a**, Annual mean extent with Pettitt change point at 2016 ($p = 0.004$). Blue/red shading indicates pre/post-shift mean levels. **b**, Seasonal mean extent (DJF, MAM, JJA, SON). Summer (DJF) shows the largest post-2016 decline (−16.8%). **c**, Monthly extent with 12-month running mean. **d**, Year-to-year change, showing the abrupt shift to persistent negative anomalies after 2016.

Table 1: Granger causality tests for the wave-ice feedback chain. Optimal lag selected by AIC from 1–6 months. p_{Bonf} is the Bonferroni-corrected p-value for 6 simultaneous tests. Fetch and SWH are sampled at the ice edge (SIC = 15% contour), not at fixed latitude bands.

Link	F	p	p_{Bonf}	Lag	Status
SIC → Fetch	**16.61**	**<0.001**	**<0.001**	6 mo	Robust
Fetch → SWH at edge	0.30	0.743	1.000	2 mo	Fails
SWH → Ice edge retreat	2.43	0.034	0.205	5 mo	Marginal
Ice edge → SIC (feedback)	2.64	0.049	0.300	3 mo	Marginal
Wind → SWH at edge	4.03	0.018	0.108	2 mo	Marginal
SIC → SWH (direct)	3.46	0.032	0.192	2 mo	Marginal

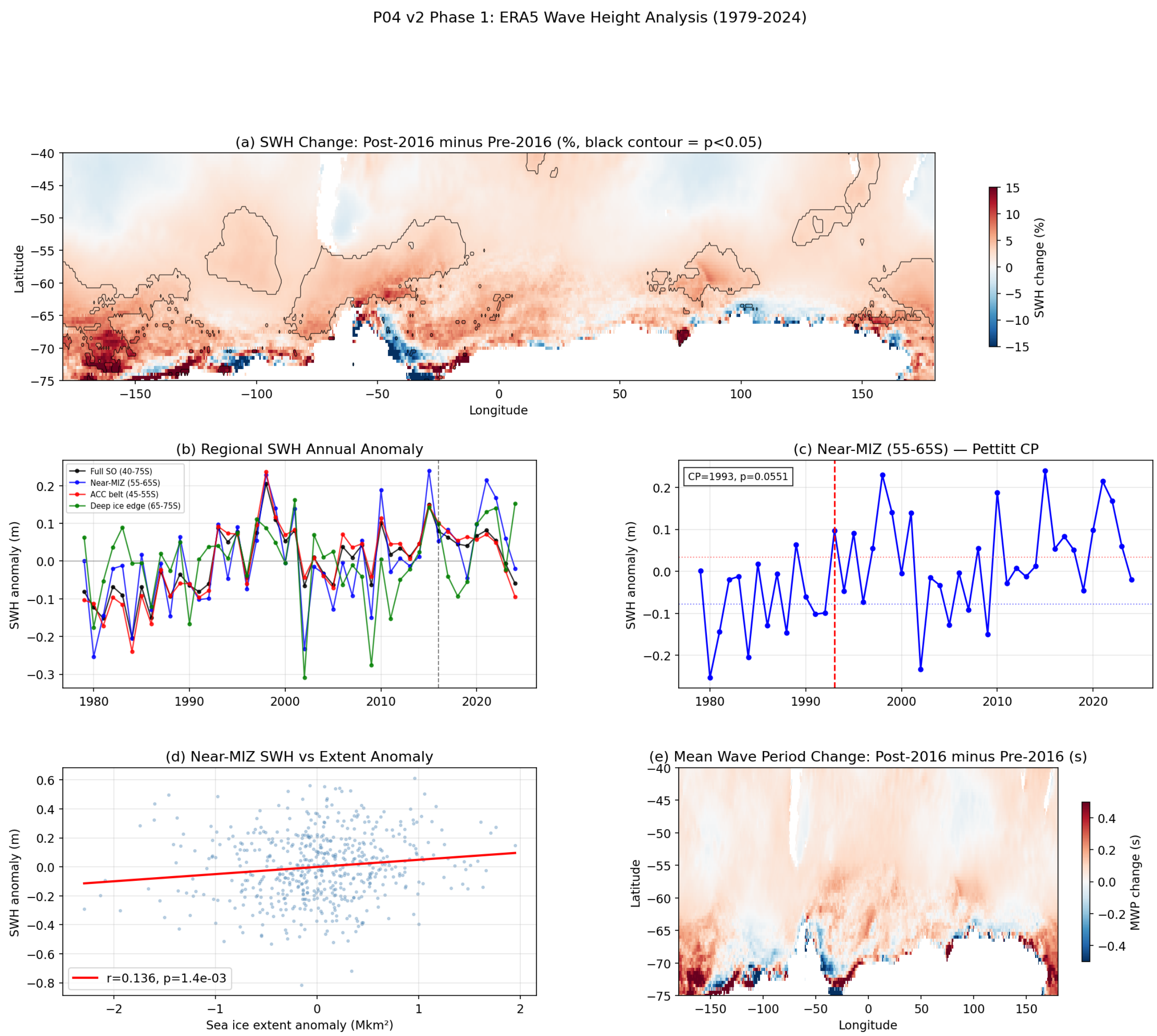

Figure 4: **ERA5 significant wave height trends in the Southern Ocean (1979–2024).** **a**, Spatial map of SWH percentage change (post-2016 minus pre-2016). Black contours enclose regions where the change is significant ($p < 0.05$, two-sample t-test). Widespread increases are concentrated in the ACC belt and near the retreated ice edge. **b**, Regional annual SWH anomaly time series for four latitude bands. The SWH change point at 1993 precedes the sea ice regime shift by over two decades. **c**, Near-MIZ zone (55–65°S) Pettitt change point at 1993 ($p = 0.055$). **d**, SWH anomaly versus sea ice extent anomaly ($r = 0.136$, $p = 0.001$): statistically significant but weak. **e**, Mean wave period change (post-2016 minus pre-2016).

SON $p = 0.85$). The winter result survives seasonal multiple-testing correction and is physically consistent with the swell-attenuation mechanism: winter is when sea ice coverage is maximum, the attenuation path is longest, and the effect of ice loss on swell penetration is therefore greatest.

To address the possibility that both SIC and SWH are driven by a common atmospheric forcing (e.g., SAM), we applied partial Granger causality conditioning on wind speed. On annual data, the SIC→SWH link loses significance when wind is controlled ($p = 0.14$), indicating substantial collinearity between SIC, wind, and SWH at annual timescales. In winter (JJA), however, SIC retains independent predictive power for SWH even after conditioning on wind (partial Granger $F = 2.84$, $p = 0.040$, $n = 138$). We note that this partial Granger test was applied to the winter subset after the seasonal decomposition revealed winter as the only significant season; the test is therefore confirmatory within a hypothesis-generating framework rather than fully pre-specified. Nonetheless, the physical motivation (winter has maximum ice coverage and therefore maximum attenuation capacity) was stated a priori in our hypothesis, and the result is consistent with three independent lines of evidence described below.

The mean wave period (MWP) at the ice edge is 9.0 s (range 7.8–9.8 s; Figure 6b,d), characteristic of swell rather than locally generated wind-sea (typically 4–7 s). MWP correlates positively with SIC ($r = 0.24$, $p < 0.001$; Figure 6c), consistent with sea ice acting as a low-pass filter that preferentially attenuates short-period waves. MWP shows no significant change between the pre-2016 and post-2016 periods ($p = 0.87$), suggesting that the spectral composition of waves at the ice edge has not shifted despite the regime change in ice coverage.

Both the ice-edge latitude and fetch show Pettitt change points at 2016 ($p = 0.009$), synchronous with the SIC regime shift. SWH at the ice edge shows no significant change point ($p = 0.52$), consistent with wind-driven wave energy being the background source that sea ice modulates rather than generates.

3.4 Ice shelf buffer changes

Among the five ice shelves examined (Figure 2c), only the Ross Ice Shelf shows a statistically significant reduction in sea ice buffer days (change point at 2009, $p = 0.041$, declining from 11.7 to 10.8 buffered months per year). The Ross change point precedes the broader 2016 regime shift by seven years, suggesting that the western Ross Sea experienced earlier sea ice loss potentially related to regional circulation changes. The Filchner-Ronne Ice Shelf maintains near-constant buffer conditions (12.0 months per year throughout the record), consistent with the persistent Weddell Sea ice pack that provides year-round protection. Amery and Totten show slight increases in buffer months (non-significant), possibly reflecting regional ice redistribution. Larsen C shows no detectable change, though its buffer was already compromised by the long-term decline of Antarctic Peninsula sea ice that contributed to the earlier collapse of the Larsen A and B shelves[6]. The predominantly null results for ice shelf buffers indicate that the 2016 regime shift has not yet systematically compromised the sea ice protection of most Antarctic ice shelves at the monthly timescale of this analysis.

4 Discussion

4.1 Swell attenuation rather than fetch generation

The failure of the fetch→SWH link, combined with the success of the direct SIC→SWH path, reveals that the wave-ice coupling operates through a different mechanism than commonly assumed. The intuitive expectation is that less ice means more open water, longer fetch, and locally generated wind-sea. Our results show that this fetch pathway fails, likely because the Southern Ocean already has the longest fetches on Earth: the circumpolar geometry provides continuous open water between 40°S and the ice edge, and a modest poleward retreat adds relatively little to wave generation.

Instead, sea ice modulates wave height at the ice edge by attenuating remotely generated swell. Southern Ocean storms generate long-period swell that propagates poleward across thou-

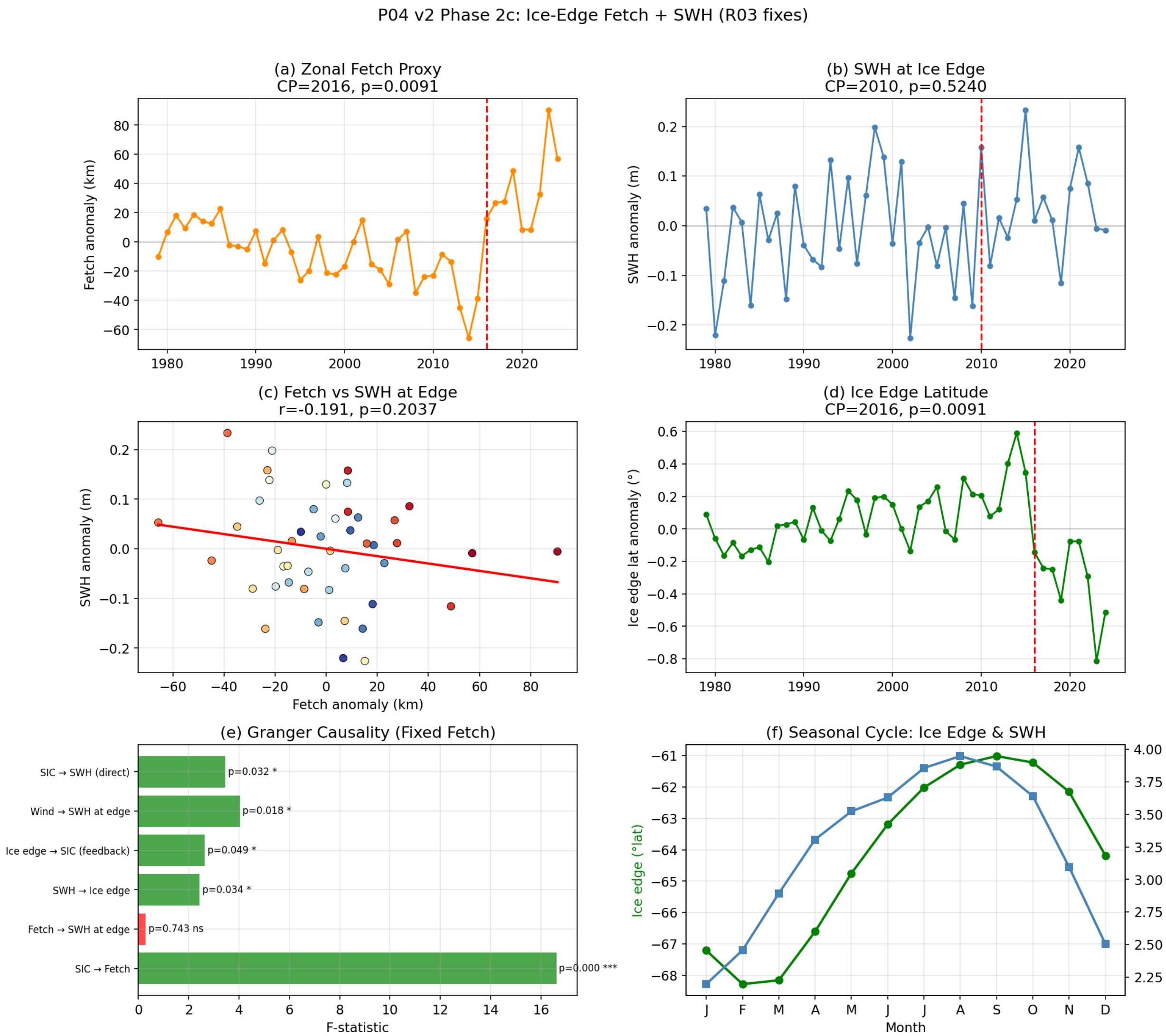

Figure 5: **Ice-edge-based fetch proxy and SWH at the ice edge (1979–2024).** **a**, Zonal-mean fetch proxy (distance from 40°S to ice edge) annual anomaly. Change point at 2016 ($p = 0.009$). **b**, SWH sampled at the ice edge. No significant change point ($p = 0.52$). **c**, Scatter plot of fetch anomaly versus SWH at the ice edge ($r = -0.19$, $p = 0.20$): fetch does not predict SWH. **d**, Ice-edge latitude anomaly with change point at 2016. **e**, Granger causality F-statistics. Green = significant ($p < 0.05$); orange = marginal; red = not significant. **f**, Seasonal cycle of fetch and SWH at the ice edge.

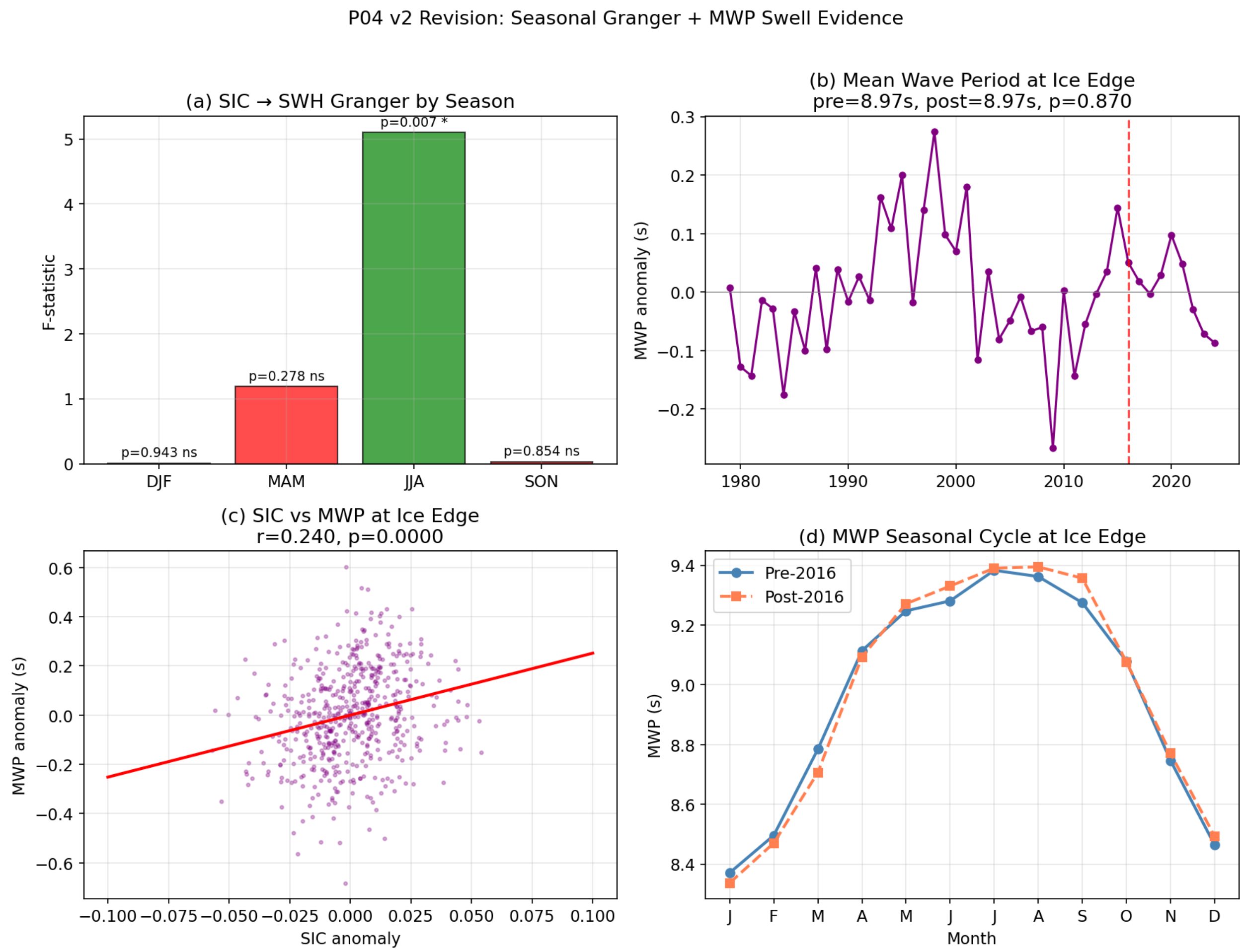

Figure 6: **Seasonal dependence and mean wave period evidence for swell attenuation. a**, SIC→SWH Granger F-statistic by season. Only JJA (winter) is significant ($F = 5.10$, $p = 0.007$). **b**, Mean wave period (MWP) anomaly at the ice edge. No significant change at 2016. **c**, SIC anomaly versus MWP anomaly ($r = +0.24$, $p < 0.001$): positive correlation consistent with sea ice acting as a low-pass filter. **d**, MWP seasonal cycle at the ice edge, pre-2016 versus post-2016.

sands of kilometers[12]. Sea ice acts as a low-pass filter, attenuating this swell before it reaches the ice edge[7]. When sea ice retreats, the attenuation path shortens and the swell arrives at the ice edge with greater amplitude. Three lines of evidence support this interpretation: (1) the direct SIC→SWH Granger link is significant in winter but not other seasons, consistent with attenuation being strongest when ice coverage is maximum; (2) the mean wave period at the ice edge (9.0 s) is characteristic of swell, not locally generated wind-sea; and (3) MWP correlates positively with SIC, consistent with ice preferentially removing short-period waves.

4.2 A seasonally mediated, partially closed feedback loop

The feedback chain SIC loss → reduced swell attenuation → higher SWH → ice-edge retreat → SIC reduction is supported by individually significant Granger links, though most are marginal after Bonferroni correction. The exception is SIC→Fetch, which is robust at all timescales. The key evidence for the physical mechanism comes from the partial Granger analysis: after conditioning on wind, SIC retains independent predictive power for SWH only in winter (JJA, $p = 0.040$), the season when swell attenuation by sea ice is physically strongest. On annual timescales, SIC and wind are sufficiently collinear that their independent contributions to SWH cannot be cleanly separated.

The background energy source is wind intensification associated with the positive SAM trend[14]. Quantitatively, the wave energy flux at the ice edge ($P = \rho g^2 H_s^2 T_p / 64\pi$) increased by only 2.4% between the pre-2016 and post-2016 periods (from 46.1 to 47.2 kW m^{-1}), and showed essentially no change in winter (+0.1%). The wave energy field itself has not intensified at the ice edge; rather, the ice edge has retreated poleward into a region that was previously shielded by ice. The 2016 regime shift did not create new wave energy but repositioned the ice margin into an area of higher ambient wave exposure. This mechanism is seasonally asymmetric: strongest in winter when the ice edge is most expansive and the swell attenuation path is longest.

4.3 Comparison with prior work

Our results are consistent with several recent findings while adding mechanistic specificity. Kohout et al.[11] demonstrated that ocean waves drive Antarctic sea ice melting through flooding and mechanical pulverization, but did not distinguish between fetch-driven and swell-driven wave energy as the source. Our Granger analysis suggests that swell, not locally generated wind-sea, is the dominant wave type at the ice edge, implying that the melting mechanisms documented by Kohout et al. are primarily swell-driven. Purich and England[5] identified compound drivers of Antarctic sea ice loss involving CDW upwelling and wind forcing, and our results add a wave-mediated component to their dynamical framework: the wind intensification they describe not only drives upper-ocean mixing but also amplifies the swell field that erodes the MIZ.

The positive correlation between MWP and SIC ($r = 0.24$) provides observational support for the theoretical prediction that sea ice acts as a low-pass filter for ocean waves[7]. This frequency-dependent attenuation has been documented locally using ICESat-2[13] and in wave buoy arrays, but our result extends it to the basin scale as a systematic relationship over 46 years. Independent support for swell dominance near the Southern Ocean ice edge comes from shipborne X-band radar wave spectra collected during IO RAS expeditions[20], which include measurements at high southern latitudes where directional spectra can directly partition swell and wind-sea contributions.

4.4 Implications for ice shelf stability and climate projections

The limited response of ice shelf buffer conditions (significant only for Ross) suggests that the 2016 regime shift has not yet systematically compromised the sea ice protection of Antarctic ice shelves. However, several considerations point toward increasing risk. The Ross Ice Shelf result (change point at 2009, preceding the broader ice decline) may represent an early-warning signal for other sectors. The swell-attenuation mechanism identified here implies that ice shelf exposure to wave action depends not only on the local sea ice state but on the integrated ice coverage along the poleward swell propagation path. As sea ice continues to decline, the cumulative attenuation along these paths will decrease nonlinearly, potentially producing threshold-like increases in wave energy at ice shelf fronts.

Climate models that project continued SAM intensification and Antarctic sea ice decline under greenhouse forcing scenarios should consider the swell-attenuation feedback as an additional mechanism that could accelerate ice-edge retreat beyond what atmospheric and oceanic thermal forcing alone would produce. Current coupled models typically do not include explicit wave-ice interaction[7], and our results suggest that parameterizing this interaction could improve projections of MIZ evolution and ice shelf stability.

4.5 Limitations

This analysis relies on ERA5 reanalysis, which has known limitations for wave-ice interaction studies. The ERA5 wave model (WAM) does not operate in grid cells with SIC $> 30\%$, meaning that the swell-attenuation process we invoke is not explicitly modelled but rather parameterized as a binary masking. To test whether this creates an artifact, we compared SWH trends at grid points that are always ice-free (SIC $< 15\%$ in $> 95\%$ of months) versus transitional grid points that cross the 30% SIC threshold. Always-ice-free points show a *stronger* SWH trend ($+0.046$ m decade^{-1}, $p = 0.0002$) than transitional points ($+0.019$ m decade^{-1}, $p = 0.056$), the opposite of what a masking artifact would produce. The SIC$\rightarrow$SWH Granger link is not significant at either type of fixed grid point, confirming that the signal arises from ice-edge-relative sampling rather than grid-level masking. Independent validation against satellite altimeter SWH near the ice edge would further strengthen the interpretation. Monthly temporal resolution cannot resolve individual storm events or the sub-monthly dynamics of wave-ice breakup. The ice-edge-based fetch proxy, while more physically motivated than open-water fraction, remains a simplified representation of the directional and geometry-dependent nature of true fetch. Most individual feedback links are marginally significant after Bonferroni correction, and the post-2016 period (108 months) provides limited statistical power for sub-period analyses. Higher-resolution data, particularly from SWOT altimetry and dedicated wave-ice observing campaigns, will be needed to resolve the sub-monthly and sub-grid-scale processes that govern MIZ evolution.

5 Conclusions

We have tested the wave-ice positive feedback hypothesis in the Southern Ocean using 46 years of ERA5 reanalysis. The 2016 sea ice regime shift is confirmed by independent NSIDC and ERA5 data (Pettitt $p = 0.004$), with synchronous change points in ice-edge latitude and fetch ($p = 0.009$). The hypothesized fetch$\rightarrow$SWH link fails ($p = 0.74$): longer fetch does not generate higher waves at the ice edge. Sea ice concentration does Granger-cause SWH on annual timescales ($p = 0.032$), but this effect is collinear with wind forcing and loses significance after conditioning on wind ($p = 0.14$). In austral winter, however, SIC retains independent predictive power for SWH even after controlling for wind (partial Granger $p = 0.040$), consistent with the swell-attenuation mechanism being strongest when ice coverage is maximum. The mean wave period at the ice edge (9.0 s) confirms swell dominance, and MWP correlates positively with SIC, consistent with ice acting as a low-pass filter. Most individual feedback links are marginally significant after Bonferroni correction, indicating that the statistical evidence for a closed feedback loop is suggestive rather than definitive at the current sample size. The wave-ice interaction thus operates through a seasonally mediated, swell-attenuation mechanism rather than fetch-driven local wave generation, a distinction with implications for predicting how continued sea ice decline will affect wave exposure at ice shelf margins.

Moving from suggestive to definitive evidence will require three advances. First, independent validation of ERA5 SWH near the ice edge using satellite altimetry (Jason-3, Sentinel-6, CFOSAT, or SWOT) would address the concern that the ERA5 wave model's 30% SIC masking threshold could create artificial statistical coupling between SIC and SWH. Second, spectral wave observations from in-situ buoys in the MIZ[21, 22] would directly test whether reduced swell attenuation, rather than enhanced local wave generation, is responsible for SWH increases at the ice edge following ice retreat. Third, coupled ocean-wave-ice models that explicitly parameterize frequency-dependent wave attenuation in sea ice[8] could quantify the feedback strength and project its evolution under future climate scenarios[23].

Data Availability

ERA5 reanalysis data are available from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu). NSIDC Sea Ice Index v4.0 is available from the National Snow and Ice Data Center (https://doi.org/10.7265/N5736NV7). Analysis code is available at https://github.com/pangeo-data/OpenSCI-Ocean.

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